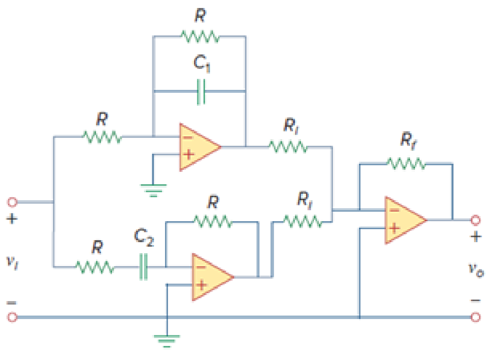
Problem

Design a notch filter based on Fig. 14.47 for $\omega 0$ = 20 krad/s, K = 5, and Q = 10. Use R = Ri = 10 kΩ.

Figure 14.47: Active band-reject filter.



Step-by-step solution

Step 1 of 7

Given that,

$$\omega_0 = 20 \text{ k rad / sec}$$

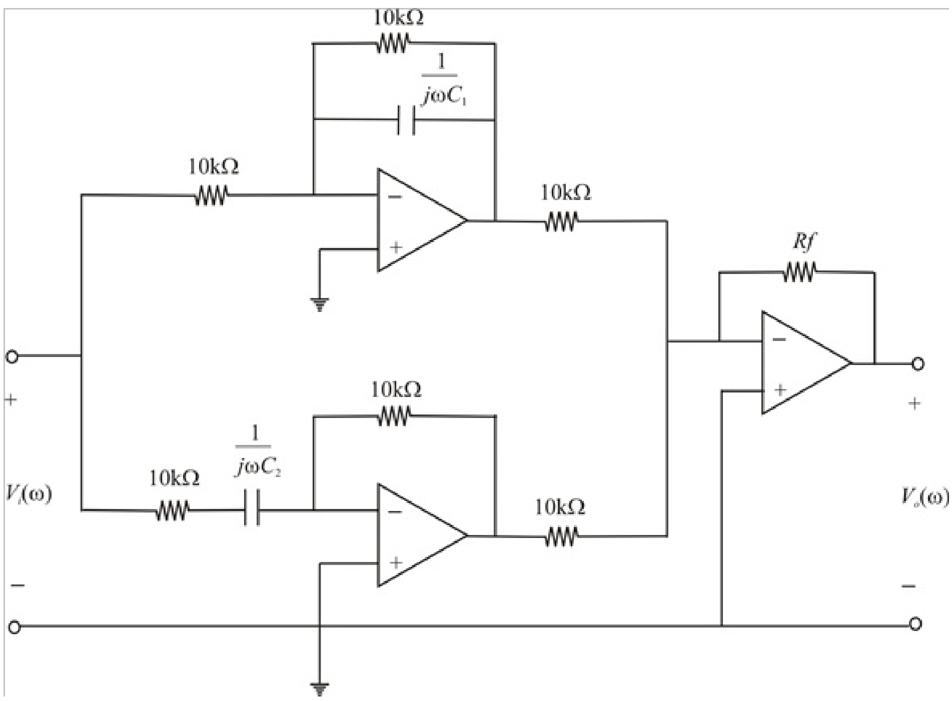
$$K = 5$$

$$Q = 10$$

$$R = R_i = 10\,\text{k}\Omega$$

Step 2 of 7

A notch filter is designed based on given values as,



Step 3 of 7

We know that,

Gain $(K) = \frac{R_f}{R_i}$

$$\Rightarrow 5 = \frac{R_f}{10\,\text{k}\Omega}$$

$$\therefore \boxed{R_f = 50\ \text{k}\Omega}$$

Step 4 of 7

Quality factor,

$$\Rightarrow Q = \frac{\omega_0}{B}$$

$$\Rightarrow B = \frac{\omega_0}{Q}$$

$$\Rightarrow B = \frac{20\times10^3}{10}$$

$$\Rightarrow B = 2000\ \text{rad /sec}$$

Step 5 of 7

Band width $(B) = \omega_2 - \omega_1$

$2000 = \omega_2 - \omega_1$ ………… (1)

Resonant frequency $(\omega_0) = \sqrt{\omega_1\omega_2}$

$$\omega_0^2 = \omega_1\omega_2$$

$$\Rightarrow (20\times10^3)^2 = \omega_1\omega_2$$

$\therefore \omega_1 = \frac{400\times10^6}{\omega_2}$ …….. (2)

Substitute value of ω_1 in equation (1), then

$$2000 = \omega_2 - \frac{(400\times10^6)}{\omega_2}$$

$$2000\omega_2 = \omega_2{}^2 - (400\times10^6)$$

$$\omega_2{}^2 - 200\omega_2 - (400\times10^6) = 0$$

By using quadratic roots expression, we get

$$\omega_2 = 21.025\text{k rad/ sec} \quad (\text{or})$$

$$\omega_2 = -19.025\text{k rad/ sec}$$

Step 6 of 7

If we take positive ω_2 and substitute in equation (2), we obtain

$$\omega_2 = \frac{1}{RC_1}$$

$$\Rightarrow C_1 = \frac{1}{\omega_2 R}$$

$$\Rightarrow C_1 = \frac{1}{\left(21.025\times10^3\right)\times\left(10\times10^3\right)}$$

$$\therefore \boxed{C_1 = 4.76\text{ nF}}$$

Step 7 of 7

While the high pas section, the lower corner frequency is,

$$\omega_1 = \frac{1}{RC_2}$$

$$\Rightarrow C_2 = \frac{1}{\omega_1 R}$$

$$\Rightarrow C_2 = \frac{1}{\left(19.025\times10^3\right)\left(10\times10^3\right)}$$

$$\therefore \boxed{C_2 = 5.26\text{ nF}}$$